

Secure File Management System with Access Control and Tamper Detection

A CAPSTONE PROJECT REPORT

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DECLARATION

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BONAFIDE CERTIFICATE

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Abstract

The Secure File Management System with Access Control and Tamper Detection is designed to provide safe, reliable, and organized storage of digital files while protecting sensitive information from unauthorized access and modification. In today's digital environment, maintaining data confidentiality, integrity, and security has become highly important for individuals and organizations. This system aims to address these challenges by integrating authentication, role-based access control, and tamper detection mechanisms into a single platform. The proposed system allows authorized users to upload, download, edit, and manage files securely. Access permissions are assigned based on user roles such as administrator and regular user, ensuring controlled access to confidential data. To enhance security, the system incorporates tamper detection techniques using file hashing methods to identify unauthorized modifications in stored files. Whenever a file is altered illegally, the system can detect the change and notify the administrator. The system also maintains logs of user activities for monitoring and accountability. Encryption techniques can be used to secure file storage and transmission, reducing the risk of data breaches. The project is developed with the objective of improving data protection, preventing unauthorized access, and ensuring file integrity in a user-friendly environment. This Secure File Management System can be applied in educational institutions, organizations, healthcare systems, and business environments where secure handling of digital files is essential. The implementation of access control and tamper detection significantly enhances the overall security and reliability of file management systems.

CHAPTER 1 – INTRODUCTION

1.1 Background of the Study

In the modern digital era, organizations and individuals store a large amount of confidential information in electronic form. Important documents such as academic records, financial files, project reports, and personal data are commonly managed using computer systems. Although digital storage provides convenience and easy accessibility, it also introduces major security concerns such as unauthorized access, data theft, and file tampering.

Traditional file management systems mainly focus on storing and retrieving files but often lack advanced security mechanisms. Files stored in unsecured systems can be modified, copied, or deleted without user permission. Such unauthorized activities may lead to data loss, privacy violations, and financial damage. Therefore, there is a growing need for secure file management systems that ensure confidentiality, integrity, and controlled access.

The Secure File Management System with Access Control and Tamper Detection is developed to address these security challenges. The system provides authentication, access restriction, file encryption, and tamper detection features using Java technologies. By combining security mechanisms with efficient file management, the proposed system helps users protect important files from unauthorized activities.

1.2 Motivation

The primary motivation behind developing this project is the increasing number of cyber threats related to digital file storage. Many users store sensitive information on computers without proper security protection, making files vulnerable to hackers and unauthorized users. Existing systems often fail to detect file modifications or provide effective monitoring mechanisms.

Another important motivation is the need to apply software engineering and Java programming concepts to solve real-world security problems. This project provides an opportunity to understand practical implementation of authentication systems, file handling, hashing algorithms, and access control mechanisms.

The project is also motivated by the importance of data integrity in modern applications. Organizations require systems that not only restrict unauthorized access but also detect if files have been altered. The integration of tamper detection improves reliability and trustworthiness of the file management system.

1.3 Overview of the Proposed System

The proposed Secure File Management System is a Java-based application designed to provide secure storage and management of files. The system allows authorized users to upload, access, download, and manage files through a secure interface.

The application uses authentication mechanisms to verify user identity before granting access. Access control policies are implemented to restrict unauthorized users from viewing or modifying files. Sensitive files are protected using encryption techniques to improve confidentiality.

A tamper detection mechanism is included to verify file integrity. The system generates hash values for files during upload and stores them securely. Whenever a file is accessed, the current hash value is compared with the original value. If a mismatch occurs, the system identifies it as tampering and alerts the user.

The proposed system improves data security, file integrity, and controlled access while maintaining user-friendly operation.

1.4 Scope of the Project

The scope of this project includes the design, development, and testing of a secure file management application using Java. The system focuses on implementing authentication, access control, encryption, and tamper detection functionalities.

The project covers:

- User registration and login
- Secure file upload and download
- File access permission management
- Hash-based tamper detection
- Basic encryption and decryption mechanisms
- Database connectivity for storing user and file details

The system is suitable for educational institutions, offices, and personal file management applications where file security is important. The project mainly focuses on software-level security and does not include advanced cloud deployment or enterprise-level distributed storage systems.

1.5 Applications of the System

The Secure File Management System can be applied in various domains where digital file security is essential.

Educational Institutions

Used for storing student records, assignments, project files, and examination documents securely.

Corporate Organizations

Helps protect confidential business files, reports, and employee records from unauthorized access.

Government Offices

Can be used for secure storage of sensitive government documents and official records.

Personal File Management

Individuals can use the system to store personal documents securely on their computers.

Research and Development

Useful for protecting research data and technical documents from tampering and unauthorized modifications.

The system demonstrates how Java-based security applications can provide practical solutions for modern data protection requirements.

Sources

Chapter 2: Problem Statement and Objectives

2.1 Problem Statement

In recent years, the rapid growth of digital technology has increased the dependency on electronic file storage systems. Organizations, educational institutions, and individuals store important information digitally for easy access and management. However, the increasing use of digital systems has also created serious security challenges such as unauthorized access, data theft, file corruption, and tampering.

Most traditional file management systems provide only basic storage and retrieval functionalities without strong security protection. Unauthorized users may gain access to confidential files and modify or delete important data. In many cases, users are unable to identify whether a file has been tampered with, resulting in loss of data integrity and trust.

Another major issue is the lack of effective access control mechanisms. Many systems do not properly restrict file access based on user roles and permissions. This creates security vulnerabilities and increases the possibility of insider attacks.

Therefore, there is a need for a secure file management solution that provides authentication, access control, encryption, and tamper detection mechanisms to ensure confidentiality, integrity, and security of digital files.

2.2 Analysis of Existing File Management Systems

Existing file management systems mainly focus on file organization and storage efficiency rather than advanced security. Most systems provide only password-based protection, which can be easily bypassed if passwords are weak or compromised.

Some cloud storage systems offer encryption features, but they may not include proper tamper detection or integrity verification. Users often depend on external antivirus software or manual monitoring to identify file modifications.

Several limitations observed in existing systems include:

- Weak authentication methods
- Lack of file integrity verification
- No tamper alert mechanism

- Poor access control management
- Increased risk of unauthorized modifications

These limitations highlight the necessity for a more secure and reliable file management system.

2.3 Need for the Proposed System

The proposed Secure File Management System is needed to overcome the limitations of traditional file storage systems. Modern applications require secure mechanisms to protect sensitive data from cyber threats and unauthorized access.

The system combines multiple security techniques such as:

- User authentication
- Role-based access control
- File encryption
- Hash-based tamper detection

By integrating these features into a single application, the proposed system improves overall data protection and file integrity. The tamper detection mechanism helps users identify unauthorized file modifications immediately.

The proposed system also reduces the dependency on external security software and provides a centralized security solution for file management.

2.4 Objectives of the Project

The main objective of this project is to develop a secure file management application using Java that protects digital files from unauthorized access and tampering.

The specific objectives are:

- To implement secure user authentication
- To provide controlled access to files

- To develop secure file upload and download mechanisms
- To implement file encryption and decryption
- To detect file tampering using hashing techniques
- To improve data confidentiality and integrity
- To create a user-friendly file management interface

The project also aims to provide practical understanding of Java security concepts and software development techniques.

2.5 Expected Outcomes

The expected outcome of the project is a fully functional Secure File Management System capable of securely storing and managing digital files.

The system is expected to:

- Prevent unauthorized file access
- Detect file tampering effectively
- Provide secure authentication and access control
- Improve reliability of digital file storage
- Enhance user confidence in data security

From an academic perspective, the project helps in understanding concepts such as file handling, database connectivity, hashing algorithms, encryption techniques, and Java GUI development.

2.6 Social and Practical Significance

The Secure File Management System has strong practical importance in today's digital environment. The project contributes to improving cybersecurity awareness and encourages secure handling of digital information.

The system is beneficial for:

- Students managing academic records
- Organizations storing confidential documents
- Government offices handling official files
- Individuals protecting personal information

The project demonstrates how software engineering and Java programming can be applied to solve real-world security problems. It also promotes the importance of data privacy, integrity, and responsible use of technology in modern society.

Chapter 3: System Architecture and Design Methodology

3.1 Introduction to System Architecture

System architecture defines the overall structure and organization of the Secure File Management System. It describes how different software components interact with each other to perform secure file storage and management operations. A proper system architecture improves reliability, security, scalability, and maintainability of the application.

The proposed system follows a modular architecture where authentication, file management, access control, encryption, and tamper detection are implemented as separate modules. This modular approach simplifies development and allows each component to function independently while contributing to the overall system performance.

The architecture is designed to ensure secure communication between users and the application while protecting files from unauthorized access and modification.

3.2 Overall System Architecture

The overall architecture of the Secure File Management System consists of the following major components:

- User Interface Module
- Authentication Module
- File Management Module
- Access Control Module
- Encryption and Decryption Module
- Tamper Detection Module
- Database Module

The user interacts with the system through a Java Swing graphical interface. The authentication module verifies user credentials before granting access to the system. Once authenticated, users can upload, download, or manage files according to

assigned permissions.

The encryption module protects files during storage, while the tamper detection module verifies file integrity using hash values. The database stores user information, file details, and hash records securely.

3.3 Block Diagram Description

The block diagram of the proposed system represents the interaction between different modules.

Working Flow:

1. User enters login credentials
2. Authentication module verifies user identity
3. Authorized users gain access to the system
4. User uploads or downloads files
5. File encryption is performed before storage
6. Hash values are generated and stored
7. During file access, integrity verification is performed
8. If tampering is detected, the user receives an alert

The block diagram demonstrates the secure flow of data between the user, application, and database.

3.4 Software Design Methodology

The software design methodology used in this project follows a modular and object-oriented approach. Java programming concepts such as classes, objects, inheritance, and exception handling are used for implementation.

The development process includes:

- Requirement analysis
- System design
- Coding and implementation
- Testing and debugging
- Deployment

Java Swing is used for developing the graphical user interface, while JDBC is used for database connectivity. The software is designed to provide simplicity, security, and efficient file handling operations.

The modular structure makes the application easier to maintain and extend in the future.

3.5 Database Design

The database is used to store user information, file details, access permissions, and hash values. MySQL database is integrated with the Java application using JDBC connectivity.

Main Tables Used:

- User Table
- File Information Table
- Access Control Table
- Hash Value Table

The database ensures secure storage and quick retrieval of information. Each uploaded file is associated with a unique hash value that helps in tamper detection.

Proper database normalization techniques are applied to avoid redundancy and improve data consistency.

3.6 Security and Access Control Design

Security is one of the most important aspects of the proposed system. The application implements multiple security mechanisms to protect sensitive files.

Authentication Security

Users must log in using valid credentials before accessing the system.

Access Control

Users are assigned permissions to restrict unauthorized operations such as viewing, editing, or deleting files.

Encryption

Files are encrypted before storage to maintain confidentiality.

Tamper Detection

SHA hashing algorithm is used to generate file hash values. During retrieval, the current hash value is compared with the original value to verify file integrity.

These security mechanisms improve reliability and reduce the risk of cyber threats.

3.7 Design Constraints and Considerations

Several constraints were considered during system design.

Performance Constraints

Large file sizes may increase processing time during encryption and hashing operations.

Security Constraints

Strong authentication and secure database handling are required to prevent attacks.

Hardware Constraints

The application should run efficiently on standard computer systems without requiring high-end hardware.

Usability Considerations

The system interface should be simple and user-friendly so that users can operate the application easily.

By considering these constraints, the system achieves better performance and usability.

3.8 Summary of Chapter

This chapter explained the architecture and design methodology of the Secure File Management System with Access Control and Tamper Detection. The modular architecture ensures secure operation and efficient management of files.

The chapter also discussed the software design process, database structure, security mechanisms, and system constraints. These design principles form the foundation for the module implementation and working process explained in the next chapter.

Chapter 4: Module Description and Working Principle

4.1 Introduction to Module-Based Design

The Secure File Management System is designed using a modular approach to improve efficiency, reliability, and maintainability. In software engineering, modular design divides the system into smaller functional units, where each module performs a specific task independently. This approach simplifies coding, testing, debugging, and future enhancements.

The proposed system consists of multiple modules such as user authentication, file management, access control, tamper detection, encryption, and database connectivity. Each module communicates with other modules to ensure secure file handling and controlled access.

The modular structure also improves scalability, allowing additional security features to be integrated in future versions of the system.

4.2 User Authentication Module

The User Authentication Module is responsible for verifying the identity of users before granting access to the system. Authentication is implemented using usernames and passwords stored securely in the database.

When a user enters login credentials, the system compares the entered details with the database records. If the credentials are valid, access is granted; otherwise, the system denies access and displays an error message.

Functions of the Authentication Module

- User registration
- User login
- Password verification
- Session management
- Unauthorized access prevention

This module improves system security by ensuring that only authorized users can access confidential files.

4.3 File Upload and Download Module

The File Management Module handles secure uploading, downloading, and storage of files. Users can select files from the system and upload them through the application interface.

During upload:

1. The selected file is encrypted
2. A hash value is generated
3. File information is stored in the database

During download:

1. User permission is verified
2. File integrity is checked
3. File is decrypted before access

Features of the Module

- File upload
- File download
- File deletion
- File viewing
- Secure file storage

This module ensures proper handling and management of files inside the system.

4.4 Access Control Module

The Access Control Module restricts unauthorized operations on files. Different users are assigned different permissions depending on their role.

Types of Access Permissions

- Read access
- Write access
- Edit access
- Delete access

Before performing any operation, the system checks whether the user has the required permission. If permission is not available, the operation is denied.

The module helps protect sensitive information from unauthorized users and reduces the possibility of data misuse.

4.5 Tamper Detection Module

The Tamper Detection Module is one of the major security features of the proposed system. It verifies whether a file has been modified after storage.

The module uses the SHA-256 hashing algorithm to generate unique hash values for files.

Working Principle

1. File is uploaded
2. Hash value is generated and stored
3. When file is accessed, a new hash value is generated
4. Both hash values are compared
5. If mismatch occurs, tampering is detected

Advantages

- Fast integrity verification
- Reliable tamper detection
- Improved file security

- Protection against unauthorized modification

This module helps maintain data integrity and reliability.

4.6 File Encryption and Decryption Module

The Encryption and Decryption Module protects files from unauthorized viewing. Before storing a file, the system encrypts its contents using encryption techniques.

Encryption Process

- Original file data is converted into unreadable format
- Encrypted file is stored securely

Decryption Process

- Authorized users request file access
- System decrypts the file
- Original content becomes accessible

Encryption improves confidentiality and prevents attackers from reading sensitive information even if they gain access to stored files.

4.7 Database Connectivity Module

The Database Connectivity Module establishes communication between the Java application and MySQL database using JDBC.

Functions of the Module

- Store user details
- Store file information
- Store hash values
- Retrieve records

- Update access permissions

The module ensures efficient data storage and retrieval operations. It also maintains consistency between the application and database.

4.8 System Integration and Working Principle

System integration combines all modules into a single secure application. The modules communicate with each other to perform file management operations efficiently.

Overall Working Principle

1. User logs into the system
2. Authentication is verified
3. User uploads or accesses files
4. Encryption and hashing are performed
5. Access permissions are checked
6. File integrity verification is completed
7. Secure access is provided to the user

This integrated approach ensures confidentiality, integrity, and controlled access throughout the system.

4.9 Reliability and Security Considerations

Reliability and security are critical aspects of the Secure File Management System. Several measures are implemented to improve system stability and protection.

Reliability Measures

- Error handling mechanisms
- Backup support

- Secure database operations
- Efficient file handling

Security Measures

- Password authentication
- Role-based access control
- File encryption
- SHA-based tamper detection

The system is designed to minimize security risks and provide dependable file management services for users.

Chapter 5: Advantages, Limitations, and Future Scope

5.1 Introduction to System Evaluation

System evaluation is an important phase in software development that helps in analyzing the performance, usability, reliability, and effectiveness of the proposed system. The Secure File Management System with Access Control and Tamper Detection is evaluated based on its security features, file handling capability, and user experience.

The proposed system is designed to improve confidentiality, integrity, and secure access to digital files. This chapter discusses the major advantages of the system, its limitations, and possible future enhancements that can further improve performance and security.

5.2 Functional Advantages of the Proposed System

The proposed system provides several functional benefits compared to traditional file management systems.

Secure User Authentication

Only authorized users can access the system using valid login credentials.

Controlled File Access

Access control mechanisms restrict unauthorized users from viewing or modifying files.

Tamper Detection

The system detects unauthorized modifications using SHA-based hash verification.

Secure File Storage

Files are encrypted before storage, improving confidentiality and protection.

User-Friendly Interface

The application provides a simple graphical interface for easy file management operations.

These features make the system more secure and reliable for handling confidential information.

5.3 Technical Advantages and Performance

From a technical perspective, the system demonstrates effective implementation of Java programming concepts and security techniques.

Advantages

- Modular architecture improves maintainability
- JDBC connectivity enables efficient database operations
- SHA-256 hashing provides reliable integrity verification
- Java Swing creates interactive graphical interfaces
- Object-oriented programming improves scalability

System Performance

The system performs file upload, download, encryption, and tamper detection operations efficiently for normal file sizes. The modular design also helps reduce software complexity and improves execution flow.

5.4 Usability and User Experience

Usability is an important factor in the success of any software application. The Secure File Management System is designed with a simple and user-friendly interface so that users can perform operations easily without technical difficulty.

User Experience Features

- Easy login and registration
- Simple file upload and download process
- Quick access to stored files

- Clear tamper detection alerts
- Easy navigation between modules

The system minimizes user effort while maintaining strong security features.

5.5 Limitations of the Proposed System

Although the proposed system provides strong security features, certain limitations still exist.

File Size Limitation

Large files may require more processing time during encryption and hashing.

Dependency on Database

The application depends on proper database connectivity for storing user and file details.

Limited Multi-User Scalability

The current version is mainly suitable for small-scale applications and may require optimization for enterprise-level usage.

Basic Security Features

Advanced cybersecurity mechanisms such as biometric authentication and intrusion detection are not included in the current implementation.

These limitations can be addressed in future versions of the project.

5.6 Security and Privacy Issues

Security and privacy are major concerns in digital file management systems. The proposed system implements several security measures to reduce risks, but proper handling of user information is still important.

Security Measures Implemented

- Password-based authentication
- File encryption
- Access permission verification
- SHA-based integrity checking

Privacy Concerns

Sensitive user information and confidential files must be stored securely to avoid unauthorized disclosure. Database security and proper password management are essential for maintaining user privacy.

Future improvements can include stronger encryption techniques and secure communication protocols.

5.7 Future Scope and Enhancement Opportunities

The Secure File Management System can be further improved by integrating advanced technologies and additional features.

Possible Future Enhancements

- Cloud storage integration
- Biometric authentication
- Two-factor authentication
- AI-based intrusion detection
- Real-time security monitoring
- Mobile application support
- Secure backup and recovery systems

The system can also be extended for enterprise-level applications with multi-user support and distributed storage mechanisms.

5.8 Social Impact and Practical Benefits

The project has practical importance in modern digital environments where secure data management is essential.

Social Benefits

- Promotes cybersecurity awareness
- Protects confidential information
- Reduces risk of unauthorized file access
- Encourages secure digital practices

Practical Applications

- Educational institutions
- Corporate organizations
- Government offices
- Personal document management

The system demonstrates how software engineering and Java programming can be used to solve real-world security problems effectively.

5.9 Summary of Chapter

This chapter presented a detailed evaluation of the Secure File Management System with Access Control and Tamper Detection. The discussion highlighted the advantages, technical performance, usability, and practical applications of the system.

The chapter also identified certain limitations and proposed future enhancements for improving security and scalability. Overall, the proposed system provides a secure and efficient solution for digital file management and integrity protection.

Chapter 6: Conclusion and Future Work

6.1 Introduction

This chapter presents the conclusion of the Secure File Management System with Access Control and Tamper Detection project. It summarizes the overall work carried out, highlights the objectives achieved, and discusses the effectiveness of the proposed system. The chapter also explains the learning outcomes, limitations faced during development, and possible future enhancements.

The conclusion chapter is important because it evaluates how successfully the project addressed the security challenges associated with digital file management systems.

6.2 Summary of the Proposed System

The Secure File Management System was developed to provide secure storage, controlled access, and tamper detection for digital files. The system integrates authentication, access control, encryption, and hashing techniques to improve confidentiality and integrity of stored data.

The project successfully demonstrated how Java technologies can be used to implement a reliable and user-friendly security application. Users can securely upload, download, and manage files while the system continuously verifies file integrity using tamper detection mechanisms.

The modular architecture and database integration improved maintainability, efficiency, and scalability of the application.

6.3 Achievement of Project Objectives

The major objectives defined at the beginning of the project were successfully achieved.

Objectives Achieved

- Secure user authentication was implemented
- Access control mechanisms restricted unauthorized operations

- Secure file upload and download operations were developed
- File encryption and decryption were implemented
- SHA-based tamper detection was successfully integrated
- User-friendly graphical interface was developed

The project achieved its goal of improving digital file security and integrity using Java programming concepts.

6.4 Performance Evaluation and Reliability

The system was tested under different operating conditions to evaluate performance and reliability.

Performance Results

- Login authentication worked accurately
- File upload and download operations were successful
- Tamper detection identified unauthorized file modifications correctly
- Database operations performed efficiently

Reliability

The system provided stable operation during testing. Error handling mechanisms improved reliability by preventing application crashes during invalid operations.

The modular structure also simplified debugging and maintenance processes.

6.5 Learning Outcomes of the Project

The project provided valuable practical knowledge and technical experience in software development and cybersecurity concepts.

Skills Learned

- Java programming and object-oriented concepts
- Java Swing GUI development
- JDBC database connectivity
- File handling techniques
- Encryption and hashing mechanisms
- Software testing and debugging

The project also improved problem-solving ability, technical documentation skills, and understanding of secure software design principles.

6.6 Limitations and Challenges Faced

During the development process, several challenges and limitations were encountered.

Challenges Faced

- Managing database connectivity issues
- Handling encryption and decryption operations
- Implementing secure file storage mechanisms
- Reducing processing time for large files

System Limitations

- Limited scalability for enterprise-level applications
- Dependency on database connectivity
- Basic security implementation compared to advanced commercial systems

Despite these limitations, the project successfully achieved its core objectives and provided effective file security mechanisms.

6.7 Scope for Future Work

The proposed system has wide scope for future improvements and advanced feature integration.

Future Enhancements

- Cloud-based secure file storage
- Two-factor authentication
- Biometric login systems
- AI-based intrusion detection
- Real-time activity monitoring
- Mobile application integration
- Automated backup and recovery systems

Future versions of the system can also support multi-user enterprise environments with advanced cybersecurity features.

6.8 Final Conclusion

In conclusion, the Secure File Management System with Access Control and Tamper Detection successfully provides a secure and efficient solution for managing digital files. The system ensures confidentiality, integrity, and controlled access through authentication, encryption, and tamper detection techniques.

The project demonstrates the practical application of Java programming and security concepts in solving real-world cybersecurity problems. With further enhancements and large-scale implementation, the system can become a more advanced and dependable platform for secure digital file management.

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APPENDICES

Appendix A – Sample Login Code

```
String username = txtUser.getText();  
  
String password = txtPass.getText();  
  
if(username.equals("admin") && password.equals("1234")){  
    System.out.println("Login Success");  
}  
else{  
    System.out.println("Invalid User");  
}
```

Appendix B – SHA Hash Code

```
import java.security.MessageDigest;  
  
MessageDigest md =  
MessageDigest.getInstance("SHA-256");
```

Appendix C – Database Tables

User Table

- User ID
- Username
- Password

File Table

- File Name
- Owner Name
- Hash Value

Appendix D – Software Requirements

- Java
- MySQL
- NetBeans/Eclipse
- JDK 8

Appendix E – Output Screens

- Login Page
- File Upload Page
- Tamper Detection Alert

Appendix F – Future Enhancements

- Cloud Storage
- Biometric Login
- Mobile App Support
- Two-Factor Authentication